

APACHE2

Web Server Hosting

Installing, Configuring & Verifying Apache2 on Kali Linux

Assignment Overview

01

Install Apache2

```
sudo apt install apache2
```

02

Check Service Status

```
sudo systemctl status apache2
```

03

Start the Service

```
sudo systemctl start apache2
```

04

Host Files

```
/var/www/html/ - projects1, sample1.html, share9.txt
```

05

Verify Hosting

```
by IP address and by localhost
```

1 Install Apache2 & Check Status

Install command:

```
sudo apt install apache2
```

Check status command:

```
sudo systemctl status apache2
```

Result: Inactive

If the service is inactive, the status output will show “Active: inactive (dead)” as in the screenshot.

```
(kali㉿kali)-[~]  
$ sudo systemctl status apache2  
o apache2.service - The Apache HTTP Server  
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)  
   Active: inactive (dead)  
     Docs: https://httpd.apache.org/docs/2.4/  
  
Jul 04 05:37:26 kali systemd[1]: Starting apache2.service - The Apache HTTP Server...  
Jul 04 05:37:26 kali apachectl[10184]: AH00558: apache2: Could not reliably determine the se  
Jul 04 05:37:26 kali systemd[1]: Started apache2.service - The Apache HTTP Server.  
Jul 04 06:13:22 kali systemd[1]: Stopping apache2.service - The Apache HTTP Server...  
Jul 04 06:13:22 kali apachectl[28611]: AH00558: apache2: Could not reliably determine the se  
Jul 04 06:13:22 kali systemd[1]: apache2.service: Killing process 11612 (apache2) with signa  
Jul 04 06:13:22 kali systemd[1]: apache2.service: Killing process 16712 (apache2) with signa  
Jul 04 06:13:22 kali systemd[1]: apache2.service: Deactivated successfully.  
Jul 04 06:13:22 kali systemd[1]: Stopped apache2.service - The Apache HTTP Server.
```

systemctl status apache2 — inactive

2

Start the Apache2 Service

Start command:

```
sudo systemctl start apache2
```

Result: Active (running)

Re-checking status confirms the service is active and running, with the process and memory details listed.

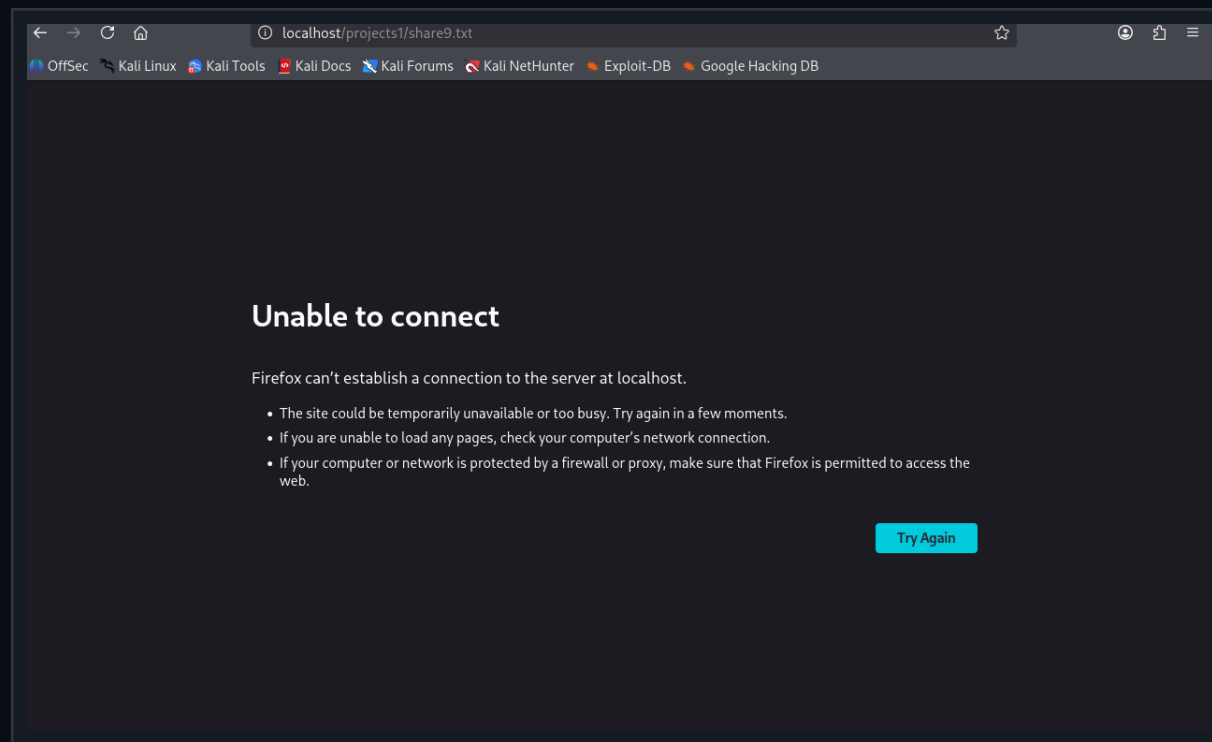
```
(kali㉿kali)-[~]  
$ sudo systemctl status apache2  
[sudo] password for kali:  
● apache2.service - The Apache HTTP Server  
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)  
   Active: active (running) since Sat 2026-07-04 05:37:26 EDT; 33min ago  
 Invocation: c58da30a5d0649aa893be4a325d7f78c  
    Docs: https://httpd.apache.org/docs/2.4/  
   Main PID: 10184 (apache2)  
   Status: "Total requests: 29; Idle/Busy workers 100/0; Requests/sec: 0.0145; Bytes served"  
   Tasks: 9 (limit: 2075)  
  Memory: 44.5M (peak: 45M)  
   CPU: 418ms  
   CGroup: /system.slice/apache2.service  
           └─10184 /usr/sbin/apache2 -k start -DFOREGROUND
```

systemctl status apache2 — active (running)

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Accessing the Server — Error Case

If the server is not active, attempting to reach it in the browser throws a connection error, as shown below.



Firefox — Unable to connect (localhost)

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Hosting Files

Web root

`/var/www/html/`

Project folder

Created `projects1` under `/var/www/html/`

Access URL

`http://localhost/projects1`

Test file 1

`sample1.html` – created inside `projects1`

Test file 2

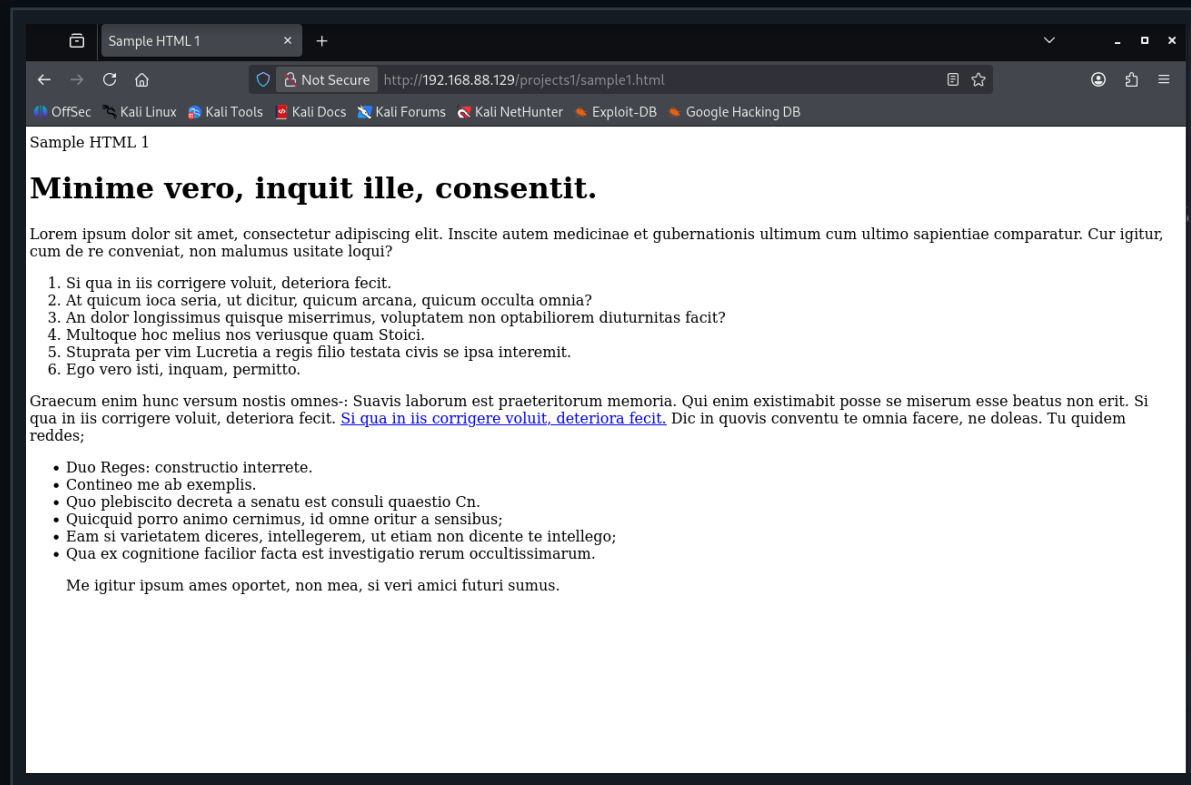
`share9.txt` – empty file created, then populated using `cat >>` and closed with `Ctrl+D`

5

Verification – Method 1: By IP Address

Accessing the hosted sample1.html page using the server's IP address confirms Apache2 is serving content correctly.

`http://192.168.88.129/projects1/sample1.html`

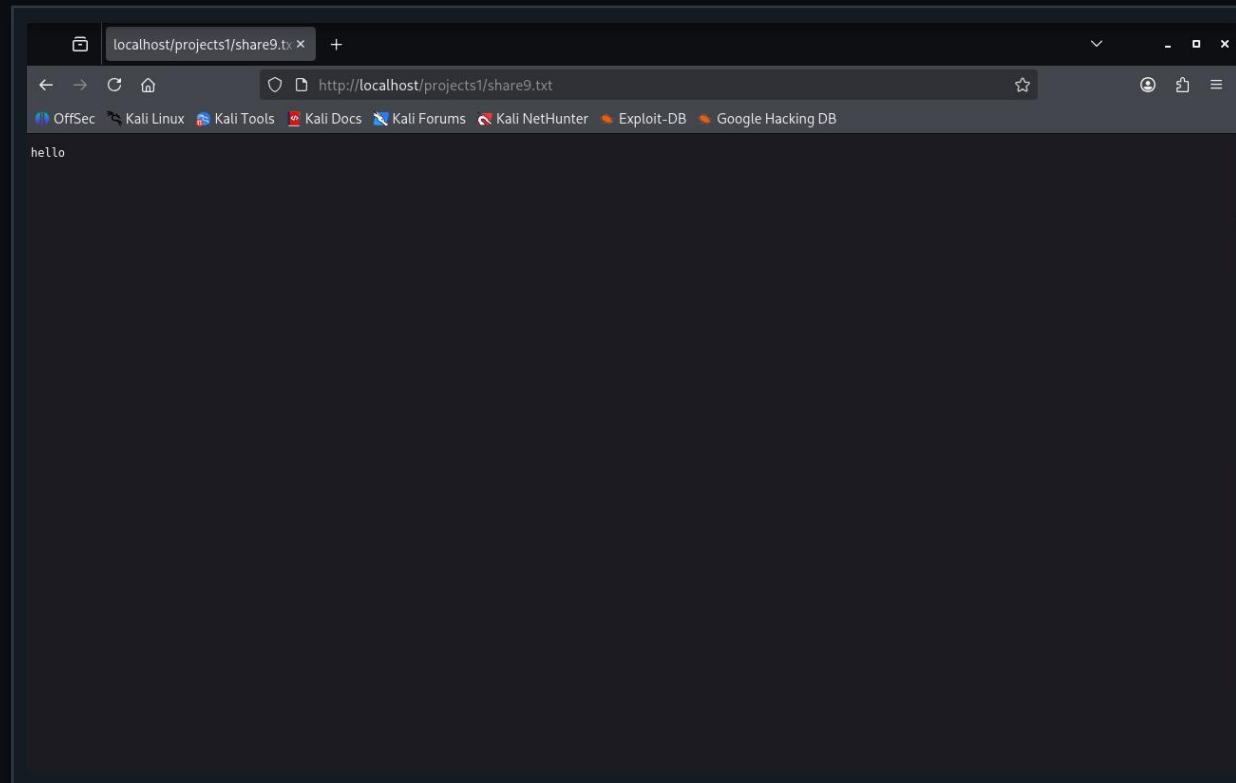


sample1.html rendered via IP address

5 Verification – Method 2: By localhost

Accessing share9.txt through localhost confirms the created file is being served from the projects1 directory.

`http://localhost/projects1/share9.txt`



share9.txt served via localhost

Summary

- › Apache2 installed with: `sudo apt install apache2`
- › Service status checked with: `sudo systemctl status apache2`
- › Inactive service started with: `sudo systemctl start apache2`
- › Files hosted from `/var/www/html/projects1` (sample1.html, share9.txt)
- › Hosting verified two ways: by IP address and by localhost